PA STEELS GRADES 6–8 · PSSA SCIENCE

Middle School Science Flashcards: Read Me First

What's included

- Grade 6: Unit 1 · Ecosystems and their Interactions (42 cards), Unit 2 · Weather and Climate Systems (38 cards), Unit 3 · Earth's Dynamic Systems (39 cards).
- Grade 7: Unit 1 · Forces and Motion (39 cards), Unit 2 · Energy (29 cards), Unit 3 · Waves and Information (28 cards), Unit 4 · Space Systems (35 cards), Unit 5 · Earth's Resources and Human Impact (31 cards), Unit 6 · Environmental Literacy and Sustainability (59 cards), Unit 7 · Engineering Design (39 cards), Unit 8 · Technology, Systems, and Society (71 cards).
- Grade 8: Unit 1 · Cells and Body Systems (34 cards), Unit 2 · Reproduction and Heredity (35 cards), Unit 3 · Natural Selection and Evolution (25 cards), Unit 4 · Properties of Matter (31 cards), Unit 5 · Chemical Reactions (24 cards).
- A blank template for making your own cards.
- Everything is in English and Spanish. Each Spanish card also shows the English term.

How to print

1. Print single-sided on regular paper (or card stock).
2. Cut on the dashed lines.
3. Fold each card on the dotted line, so the term is on the front and the answer is on the back. A little glue or tape keeps it closed.

How to study

- Look at the front, say or write the answer from memory, then check.
- Put cards you missed in a "still learning" pile and practice them again.
- A little each day (10–15 minutes) works better than one long session.

Standards key

Each card shows the PA STEELS standard it supports. Together, the cards cover all 102 grades 6–8 standards tested on the PSSA Grade 8 Science test, in four areas: Life Science with Environmental Literacy and Sustainability, Physical Science, Earth and Space Science, and Technology and Engineering. Grades 6 and 8 follow the Wilson School District science units. Wilson's 7th grade units are not posted online, so the 7th grade units are an estimate.

Standard	Students who demonstrate understanding can...
Grade 6 · Unit 1 · Ecosystems and their Interactions	
3.1.6-8.F	construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms.
3.1.6-8.I	analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem.
3.1.6-8.J	construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems.
3.1.6-8.K	develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem.
3.1.6-8.L	construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.
3.1.6-8.U	evaluate competing design solutions for maintaining biodiversity and ecosystem services.
3.4.6-8.C	develop a model to describe how watersheds and wetlands function as systems, including the roles and functions they serve.
Grade 6 · Unit 2 · Weather and Climate Systems	
3.3.6-8.H	develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity.
3.3.6-8.I	develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates.
3.3.6-8.J	collect data to provide evidence for how the motion and complex interactions of air masses result in changes in weather conditions.
3.3.6-8.O	ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.
Grade 6 · Unit 3 · Earth's Dynamic Systems	
3.3.6-8.D	construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's 4.6-billion-year-old history.
3.3.6-8.E	construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales.
3.3.6-8.F	develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process.
3.3.6-8.G	analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of the past plate motions.
3.3.6-8.L	analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects.

Standard	Students who demonstrate understanding can...
Grade 7 · Unit 1 · Forces and Motion	
3.2.6-8.G	apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects.
3.2.6-8.H	plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object.
3.2.6-8.I	ask questions about data to determine the factors that affect the strength of electric and magnetic forces.
3.2.6-8.J	construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects.
3.2.6-8.K	conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects exerting forces on each other even though the objects are not in contact.
Grade 7 · Unit 2 · Energy	
3.2.6-8.L	construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass and to the speed of an object.
3.2.6-8.M	apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer.
3.2.6-8.N	plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample.
3.2.6-8.O	construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object.
3.2.6-8.P	develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system.
Grade 7 · Unit 3 · Waves and Information	
3.2.6-8.Q	use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave.
3.2.6-8.R	develop and use a model to describe that waves are reflected, absorbed, or transmitted through various materials.
3.2.6-8.S	integrate qualitative scientific and technical information to support the claim that digitized signals are a more reliable way to encode and transmit information than analog signals.
Grade 7 · Unit 4 · Space Systems	
3.3.6-8.A	develop and use a model of the Earth-sun-moon system to describe the cyclic patterns of lunar phases, eclipses of the sun and moon, and seasons.
3.3.6-8.B	develop and use a model to describe the role of gravity in the motions within galaxies and the solar system.
3.3.6-8.C	analyze and interpret data to determine scale properties of objects in the solar system.

Standard	Students who demonstrate understanding can...
Grade 7 · Unit 5 · Earth's Resources and Human Impact	
3.3.6-8.K	construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes.
3.3.6-8.M	apply scientific principles to design a method for monitoring and minimizing human impact on the environment.
3.3.6-8.N	construct an argument supported by evidence for how increases in human population and per capita consumption of natural resources impact Earth's systems.
Grade 7 · Unit 6 · Environmental Literacy and Sustainability	
3.4.6-8.A	develop a model to describe how agricultural and food systems function, including the sustainable use of natural resources and the production, processing, and management of food, fiber, and energy.
3.4.6-8.B	analyze and interpret data about how different societies (economic and social systems) and cultures use and manage natural resources differently.
3.4.6-8.D	gather, read, and synthesize information from multiple sources to investigate how Pennsylvania environmental issues affect Pennsylvania’s human and natural systems.
3.4.6-8.E	collect, analyze, and interpret environmental data to describe a local environment.
3.4.6-8.F	obtain and communicate information on how integrated pest management could improve indoor and outdoor environments.
3.4.6-8.G	obtain and communicate information to describe how best resource management practices and environmental laws are designed to achieve environmental sustainability.
3.4.6-8.H	design a solution to an environmental issue in which individuals and societies can engage as stewards of the environment.
3.4.6-8.I	construct an explanation that describes regional environmental conditions and their implications on environmental justice and social equity.

Standard	Students who demonstrate understanding can...
Grade 7 · Unit 7 · Engineering Design	
3.5.6-8.DD	engage in a research and development process to simulate how inventions and innovations have evolved through systematic tests and refinements.
3.5.6-8.II	predict outcomes of a future product or system at the beginning of the design process.
3.5.6-8.JJ	apply informed problem-solving strategies to the improvement of existing devices or processes or the development of new approaches.
3.5.6-8.M	develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.
3.5.6-8.N	analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.
3.5.6-8.O	interpret the accuracy of information collected.
3.5.6-8.P	evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.
3.5.6-8.Q	apply a technology and engineering design thinking process.
3.5.6-8.R	develop innovative products and systems that solve problems and extend capabilities based on individual or collective needs and wants.
3.5.6-8.S	illustrate the benefits and opportunities associated with different approaches to design.
3.5.6-8.T	create solutions to problems by identifying and applying human factors in design.
3.5.6-8.U	evaluate and assess the strengths and weaknesses of various design solutions given established principles and elements of design.
3.5.6-8.V	refine design solutions to address criteria and constraints.
3.5.6-8.W	define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.
3.5.6-8.X	defend decisions related to a design problem.

Standard	Students who demonstrate understanding can...
Grade 7 · Unit 8 · Technology, Systems, and Society	
3.5.6-8.A	research information from various sources to use and maintain technological products or systems.
3.5.6-8.AA	adapt and apply an existing product, system, or process to solve a problem in a different setting.
3.5.6-8.B	use instruments to gather data on the performance of everyday products.
3.5.6-8.BB	demonstrate how knowledge gained from other content areas affects the development of technological products and systems.
3.5.6-8.C	hypothesize what alternative outcomes (individual, cultural, and/or environmental) might have resulted had a different technological solution been selected.
3.5.6-8.CC	consider historical factors that have contributed to the development of technologies and human progress.
3.5.6-8.D	analyze how the creation and use of technologies consumes renewable, non-renewable, and inexhaustible resources; creates waste; and may contribute to environmental challenges.
3.5.6-8.E	consider the impacts of a proposed or existing technology and devise strategies for reducing, reusing, and recycling waste caused by its creation.
3.5.6-8.EE	differentiate between inputs, processes, outputs, and feedback in technological systems.
3.5.6-8.F	analyze examples of technologies that have changed the way people think, interact, live, and communicate.
3.5.6-8.FF	demonstrate how systems thinking involves considering relationships between every part, as well as how the systems interact with the environment in which it is used.
3.5.6-8.G	analyze how an invention or innovation was influenced by the context and circumstances in which it is developed.
3.5.6-8.GG	create an open-loop system that has no feedback path and requires human intervention.
3.5.6-8.H	evaluate trade-offs based on various perspectives as part of a decision process that recognizes the need for careful compromises among competing factors.
3.5.6-8.HH	create a closed-loop system that has a feedback path and requires no human intervention.
3.5.6-8.I	examine the ways that technology can have both positive and negative effects at the same time.
3.5.6-8.J	use tools, materials, and machines to safely diagnose, adjust, and repair systems.
3.5.6-8.K	use devices to control technological systems.
3.5.6-8.KK	explain how technology and engineering are closely linked to creativity, which can result in both intended and unintended innovations.
3.5.6-8.L	design methods to gather data about technological systems.
3.5.6-8.LL	compare how different technologies involve different sets of processes.
3.5.6-8.Y	compare, contrast, and identify overlap between the contributions of science, technology, engineering, and mathematics in the development of technological systems.
3.5.6-8.Z	analyze how different technological systems often interact with economic, environmental, and social systems.

Standard	Students who demonstrate understanding can...
Grade 8 · Unit 1 · Cells and Body Systems	
3.1.6-8.A	conduct an investigation to provide evidence that living things are made of cells; either one cell or many different numbers and types of cells.
3.1.6-8.B	develop and use a model to describe the function of a cell as a whole and the ways the parts of cells contribute to the function.
3.1.6-8.C	use arguments supported by evidence for how the body is a system of interacting subsystems composed of groups of cells.
3.1.6-8.G	develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism.
3.1.6-8.H	gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories.
Grade 8 · Unit 2 · Reproduction and Heredity	
3.1.6-8.D	use arguments based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants, respectively.
3.1.6-8.E	construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms.
3.1.6-8.M	develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral effects to the structure and function of the organism.
3.1.6-8.N	develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation.
3.1.6-8.R	gather and synthesize information about the technologies that have changed the way humans influence the inheritance of desired traits in organisms.
Grade 8 · Unit 3 · Natural Selection and Evolution	
3.1.6-8.O	analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth under the assumption that natural laws operate today as in the past.
3.1.6-8.P	apply scientific ideas to construct an explanation for the anatomical similarities and differences among modern organisms and between modern and fossil organisms to infer evolutionary relationships.
3.1.6-8.Q	analyze displays of pictorial data to compare patterns of similarities in anatomical structures across multiple species to identify relationships not evident in the fully formed anatomy.
3.1.6-8.S	construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment.
3.1.6-8.T	use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

Standard	Students who demonstrate understanding can...
Grade 8 · Unit 4 · Properties of Matter	
3.2.6-8.A	develop models to describe the atomic composition of simple molecules and extended structures.
3.2.6-8.B	develop a model that predicts and describes changes in particle motion, temperature and state of a pure substance when thermal energy is added or removed.
3.2.6-8.C	gather and make sense of information to describe that synthetic materials come from natural resources and impact society.
Grade 8 · Unit 5 · Chemical Reactions	
3.2.6-8.D	analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred.
3.2.6-8.E	develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.
3.2.6-8.F	undertake a design project to construct, test, and modify a device that either releases or absorbs thermal energy by chemical processes.

Standard wording is from the Pennsylvania Department of Education STEELS Foundation Boxes (updated September 2025). The four test areas and 102 standards are from PDE's 2025 PSSA Science Test Design.